

LESSON 4

Discrete probability distributions

1. Bernoulli distribution

- 1.1. Definition
- 1.2. Mathematical expectation

2. Binomial distribution

- 2.1. Definition
- 2.2. Mathematical expectation
- 2.3 Distribution table

3. Poisson distribution

- 3.1. Definition
- 3.2. Mathematical expectation
- 3.3. Poisson's abacus and table

4. Inspection plans

- 4.1. O-C Curve

Bernoulli distribution $Be(p)$ - Definition

To conduct a random experiment **once** and check whether a certain **event occurs or not**. This experiment can be defined as a dichotomous or Bernoulli variable:

- $Be(p) \sim X$ where $X = \begin{cases} 0 & \text{if } X \text{ does not occur} \\ 1 & \text{if } X \text{ occurs} \end{cases}$
- p : probability of the event occurring
- $q = (1-p)$: probability of the event not occurring

Be(p) – Mathematical expectation

Mean:

$$E(X) = p$$

Variance:

$$\sigma^2(X) = p(1 - p)$$

Standard deviation:

$$\sigma(X) = \sqrt{p(1 - p)}$$

Binomial distribution $Bi(n,p)$ - Definition

Given a binary test, with **two possible outcomes**, identified as success and failure, variable X is studied: number of successes **x achieved in n independent trials** of this test.

- $Bi(n, p) \sim X$ where $X = [0, 1, \dots, n]$
- p : probability of the event occurring. All tests have the same chance of success.
- $q = (1-p)$: probability of the event not occurring

Binomial distribution $Bi(n,p)$ - Definition

Probability function:

$$P(X = x) = \binom{n}{x} p^x \cdot (1 - p)^{n-x}$$

A Bernoulli distribution is equal to a Binomial distribution when $n=1$:

$$Bi(1, p) = Be(p)$$

$$Bi(n, p) = Be_1(p) + Be_2(p) + \cdots + Be_n(p)$$

Bi(n,p) – Mathematical expectation

Mean:

$$E(X) = n \cdot p$$

Variance:

$$\sigma^2(X) = n \cdot p(1 - p)$$

Standard deviation:

$$\sigma(X) = \sqrt{n \cdot p(1 - p)}$$

Binomial's Table

Calculates probability $P(X = x)$ and distribution functions $F(X) = P(X \leq x)$

x : number of times that happens the event

n : number of trials

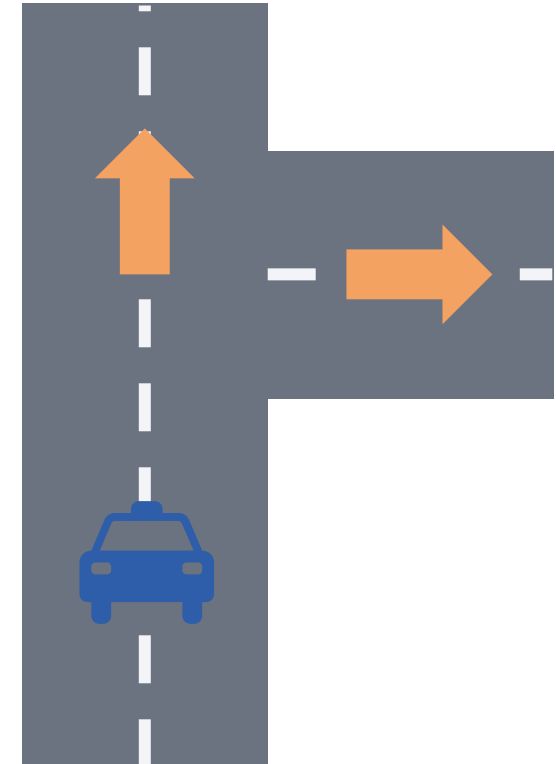
n	x	P(X=x)					P(X≤x)				
		p = 0,1	p = 0,2	p = 0,3	p = 0,4	p = 0,5	p = 0,1	p = 0,2	p = 0,3	p = 0,4	P = 0,5
n=5	0	0,5900	0,3280	0,1680	0,0780	0,0310	0,5900	0,3280	0,1680	0,0780	0,0310
	1	0,3280	0,4100	0,3600	0,2590	0,1560	0,9190	0,7370	0,5280	0,3370	0,1880
	2	0,0730	0,2050	0,3090	0,3460	0,3130	0,9910	0,9420	0,8370	0,6830	0,5000
	3	0,0080	0,0510	0,1320	0,2300	0,3130	1,0000	0,9930	0,9690	0,9130	0,8130
	4	0,0000	0,0060	0,0280	0,0770	0,1560	1,0000	1,0000	0,9980	0,9900	0,9690
	5	0,0000	0,0000	0,0020	0,0100	0,0310	1,0000	1,0000	1,0000	1,0000	1,0000
n=10	0	0,3490	0,1070	0,0280	0,0060	0,0010	0,3490	0,1070	0,0280	0,0060	0,0010
	1	0,3870	0,2680	0,1210	0,0400	0,0100	0,7360	0,3760	0,1490	0,0460	0,0110
	2	0,1940	0,3020	0,2330	0,1210	0,0440	0,9300	0,6780	0,3830	0,1670	0,0550
	3	0,0570	0,2010	0,2670	0,2150	0,1170	0,9870	0,8790	0,6500	0,3820	0,1720
	4	0,0110	0,0880	0,2000	0,2510	0,2050	0,9980	0,9670	0,8500	0,6330	0,3770
	5	0,0010	0,0260	0,1030	0,2010	0,2460	1,0000	0,9940	0,9530	0,8340	0,6230
	6	0,0000	0,0060	0,0370	0,1110	0,2050	1,0000	0,9990	0,9890	0,9450	0,8280
	7	0,0000	0,0010	0,0090	0,0420	0,1170	1,0000	1,0000	0,9980	0,9880	0,9450
	8	0,0000	0,0000	0,0010	0,0110	0,0440	1,0000	1,0000	1,0000	0,9980	0,9890
	9	0,0000	0,0000	0,0000	0,0020	0,0100	1,0000	1,0000	1,0000	1,0000	0,9990
	10	0,0000	0,0000	0,0000	0,0000	0,0010	1,0000	1,0000	1,0000	1,0000	1,0000

P: pronbability for distribution function

p: pronbability for probability function

Bi(n,p) – Exercise 1

Determine the probability that, out of a series of 6 vehicles waiting for a traffic light to turn green, 3 will turn right, knowing that they can only choose between going straight or turning right and that the probability of the latter decision is 0.1.



Bi(n,p) – Exercise 2

A spare parts warehouse receives a batch of thousands of rivets, 1% of which are defective.

- a) What is the probability that a random sample of 50 rivets from that batch will contain 2 defective rivets?
- b) What is the expected number of defective rivets?
- c) For the acceptance of received batches, a quality control plan is applied that consists of randomly sampling 50 rivets and accepting the batch if a maximum of 2 defective rivets appear. What is the probability of accepting the aforementioned batch with this control plan?

Poisson distribution $P(\lambda)$ - Definition

Given a random experiment, the variable indicates the number of occurrences (x) of an event **in a given space or time**.

- $P_s(\lambda) \sim X$ where λ represents the probability of an occurrence
- The **probability** of an occurrence is **identical for intervals of equal length**.
- The potential number of occurrences is theoretically unlimited: $x \rightarrow \infty$
- The number of occurrences in separate intervals is **statistically independent**.

Poisson distribution $P(\lambda)$ - Definition

Probability function:

$$P(X = x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!}$$

The Poisson distribution is the limit to which the distribution of a binomial variable tends:

$$\text{When } n \rightarrow \infty \text{ and } p \rightarrow 0 \text{ then } n \cdot p \rightarrow \lambda$$

To make a more accurate approximation of the Poisson distribution:

$$n \geq 50 \text{ and } p \leq 0.01$$

$P(\lambda)$ – Mathematical expectation

Mean:

$$E(X) = \lambda$$

Variance:

$$\sigma^2(X) = \lambda$$

Standard deviation:

$$\sigma(X) = \sqrt{\lambda}$$

Poisson's Table

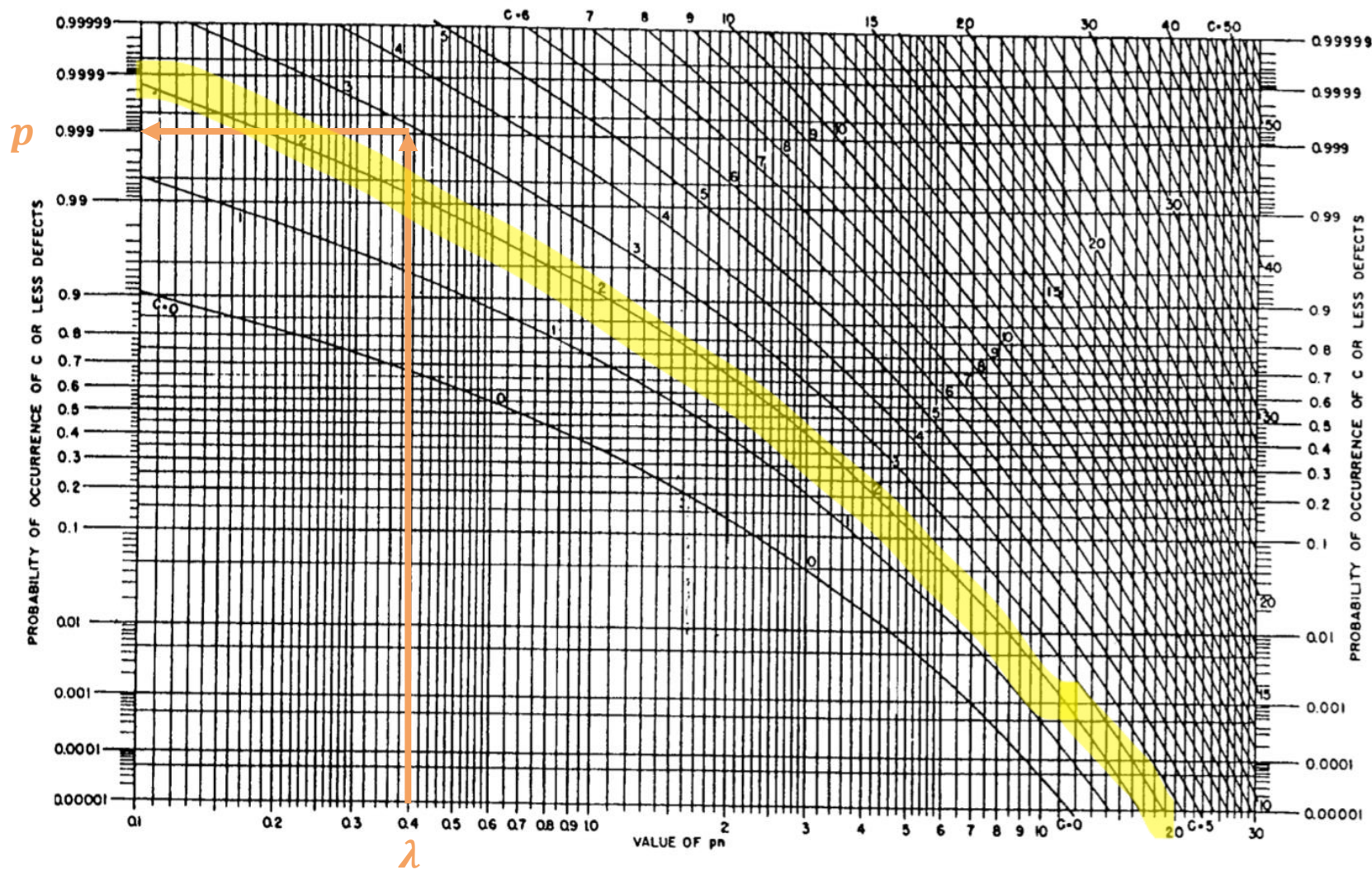
Calculates distribution functions $F(X) = P(X \leq x) = \sum_{i=1}^x e^{-\lambda} \frac{\lambda^x}{x!}$

λ : lambda
parameter for
Poisson
distribution

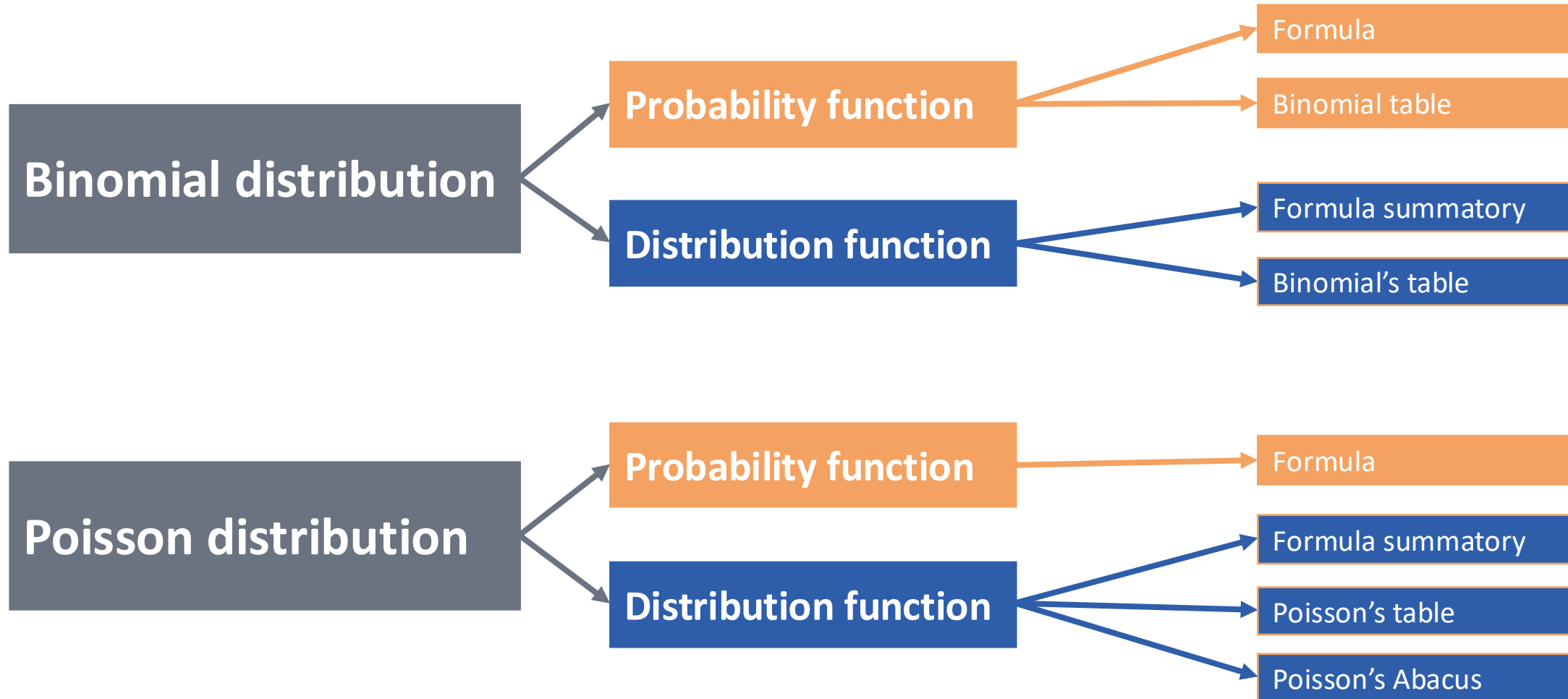
λ	c=0	c=1	c=2	c=3	c=4	c=5	c=6	c=7	c=8
0,02	0,9802	0,9998	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000
0,04	0,9608	0,9992	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000
0,06	0,9418	0,9983	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000
0,08	0,9231	0,9970	0,9999	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000
0,10	0,9048	0,9953	0,9998	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000
0,15	0,8607	0,9898	0,9995	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000
0,20	0,8187	0,9825	0,9989	0,9999	1,0000	1,0000	1,0000	1,0000	1,0000
0,25	0,7788	0,9735	0,9978	0,9999	1,0000	1,0000	1,0000	1,0000	1,0000
0,30	0,7408	0,9631	0,9964	0,9997	1,0000	1,0000	1,0000	1,0000	1,0000
0,35	0,7047	0,9513	0,9945	0,9995	1,0000	1,0000	1,0000	1,0000	1,0000
0,40	0,6703	0,9384	0,9921	0,9992	0,9999	1,0000	1,0000	1,0000	1,0000
0,45	0,6376	0,9246	0,9891	0,9988	0,9999	1,0000	1,0000	1,0000	1,0000
0,50	0,6065	0,9098	0,9856	0,9982	0,9998	1,0000	1,0000	1,0000	1,0000
0,55	0,5769	0,8943	0,9815	0,9975	0,9997	1,0000	1,0000	1,0000	1,0000

c=x: number of
times that
happens the
event

Poisson's Abacus



Calculations summary



$P(\lambda)$ – Exercise 3

A spare parts warehouse receives a batch of thousands of rivets, 1% of which are defective. **Calculate using Poisson distribution:**

- a) What is the probability that a random sample of 50 rivets from that batch will contain 2 defective rivets?
- b) What is the expected number of defective rivets?
- c) For the acceptance of received batches, a quality control plan is applied that consists of randomly sampling 50 rivets and accepting the batch if a maximum of 2 defective rivets appear. What is the probability of accepting the aforementioned batch with this control plan?

$P(\lambda)$ – Exercise 4

The number of paint defects on the casing of a washing machine can be assumed to be a random variable with a Poisson distribution with a mean of $\lambda=2$ defects per unit.

- a) Determine the probability that, when randomly selecting a casing from that production run, exactly 3 defects will appear on it.
- b) To control the quality of the manufacturing process for the casings, one of the manufactured casings is taken at random and the process is accepted as correct if 2 or fewer defects are found.
- c) What is the probability of accepting as correct, with this control, a process in which an average of 5 defects per unit manufactured occur?

$P(\lambda)$ – Exercise 5

Knowing that on average there are 3 fatal accidents per day on Spanish roads (excluding weekends):

- a) What is the probability that there will be at least one fatal accident next Tuesday?
- b) What is the probability that there will be more than 7 fatal accidents next Tuesday?
- c) What is the probability that there will be exactly 7 fatal accidents next Tuesday?

Assuming that the variables associated with the different days of the week are independent and that the average number of fatal accidents at the weekend (Saturday plus Sunday) is equal to 15:

- d) Calculate the probability that there will be fewer than 25 fatal accidents next week.
- e) Calculate the probability that there will be fewer than 25 fatal accidents next week if we know that there have already been 8 fatal accidents during the first 2 days.
- f) What would be the probability that there will be fewer than 25 fatal accidents in each of two consecutive weeks?
- g) What would be the probability that during two weeks, in one of them there will be fewer than 25 fatal accidents?

Inspection plans

X: number of defective parts.

When establishing an inspection plan, it is necessary to set two parameters:

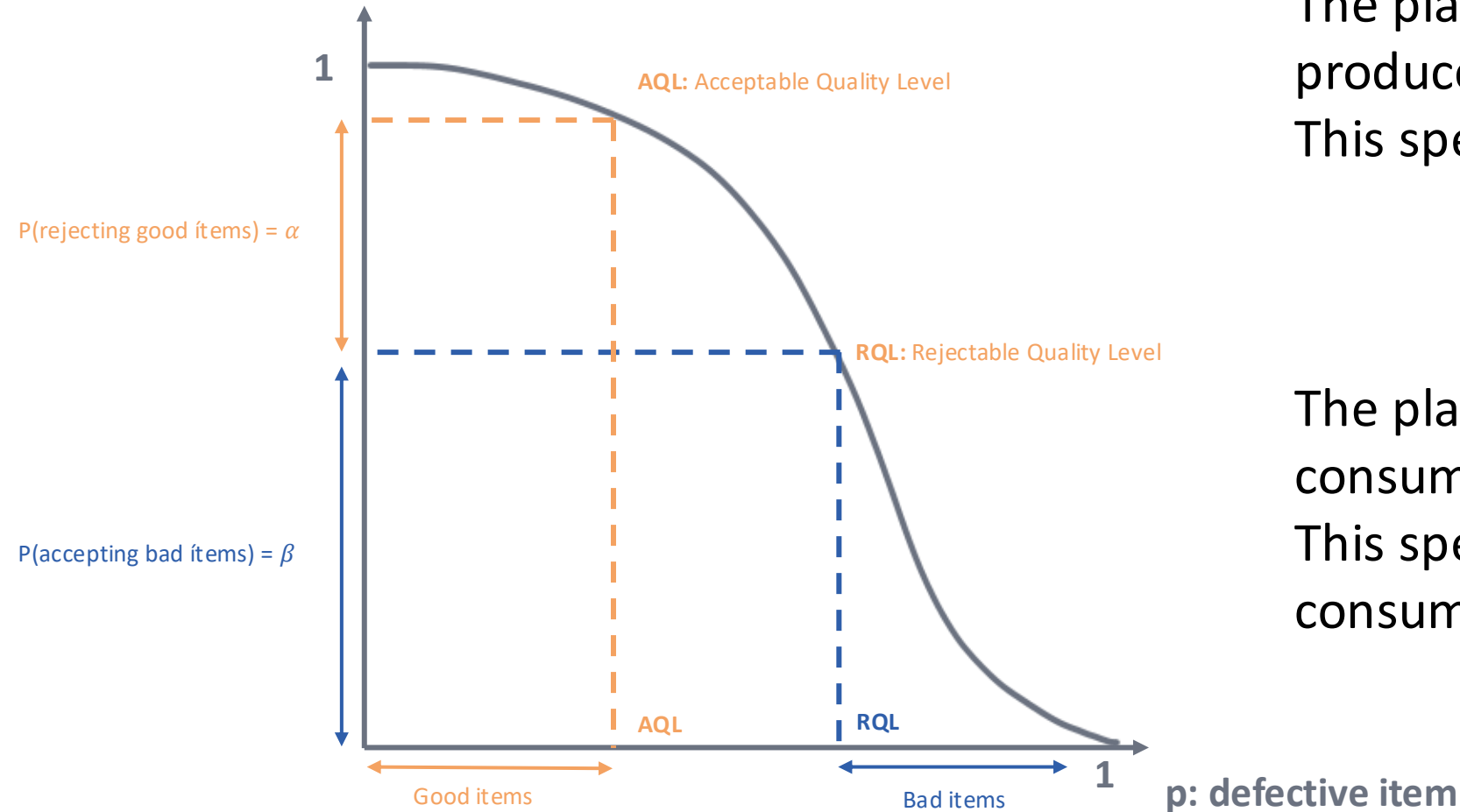
- **n**: sample size
- **c**: maximum number of defective parts in the sample to accept the batch.

Acceptance criteria:

- *If $X > c \rightarrow$ The item is rejected*
- *If $X \leq c \rightarrow$ The item is accepted*

O-C Curve

$P(\text{accepting item} | p)$



The plan will be rejected by the producer with a small probability α . This specifies the producer's risk:

$$P(X \leq AQL) = 1 - \alpha$$

The plan will be accepted by the consumer with a small probability β . This specifies the risk to the consumer:

$$P(X \geq RQL) = \beta$$

Inspection plans – Exercise 6

To control the quality of a batch of tires received at a car factory, the following two-stage sampling plan is used:

A first sample of 50 tires is taken, accepting the batch if all tires are correct and rejecting it if there are more than two defective tires

In all other cases, a new sample of another 50 tires is taken, and the batch is definitively accepted if there are fewer than 3 defective tires in the total of the two samples, and rejected if there are more.

- a) What is the probability that the above plan will accept a batch that has 1% defective tires?
- b) If all batches arriving at the factory have 1% defective tires, on average how many tires will be examined per batch?